

The Guardian – Scam & Phishing Detection Module

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Evaluation and Validation Report

This report presents the evaluation results of the Scam & Phishing Detection Module developed as part of The Guardian system. The module is designed to identify social engineering scams, phishing attempts, and malicious links in user-generated text on social media platforms. The detection system combines URL analysis and AI-based linguistic pattern recognition to identify scam messages in real time and warn users before they interact with potentially dangerous content.

1. Module Architecture Overview

The Scam & Phishing Detection Module is designed to detect social engineering scams, phishing attempts, and malicious links within user-generated text on social media platforms. The module uses a hybrid detection architecture combining URL analysis and AI-based language classification.

The processing pipeline follows these stages:

- 1. URL Extraction and Analysis** – URLs are extracted using pattern matching and checked against suspicious domain patterns.
- 2. Text Preprocessing** – The system normalizes slang, corrects spelling, and removes noise to improve model understanding.
- 3. Scam Language Detection** – A fine-tuned DistilBERT NLP model analyzes the surrounding text.
- 4. Social Engineering Pattern Recognition** – The system detects linguistic indicators such as false urgency, impersonation, financial reward claims, and suspicious call-to-action phrases.
- 5. Risk Scoring** – URL analysis and language model outputs are combined to generate a final scam probability score.
- 6. User Alert Generation** – If the risk score exceeds the threshold, the browser extension displays a warning to the user.

2. Dataset for Evaluation

A labeled evaluation dataset consisting of 5,683 social-media style messages was used to validate the module's performance.

The dataset includes:

- Legitimate social media posts and comments
- Known phishing messages
- Scam advertisements and impersonation attempts
- Messages containing malicious or suspicious links

Each message was labeled with a binary classification label:

Safe (0) – Legitimate content

Scam (1) – Scam or phishing message

3. Evaluation Methodology

The module's predictions were compared against the labeled ground truth dataset using standard classification metrics.

The following evaluation metrics were computed:

- Precision
- Recall
- F1-score
- Accuracy
- Confusion Matrix

Precision measures how many flagged scam messages were correct, while recall measures how many actual scams were successfully detected.

The F1-score provides a balanced evaluation combining both precision and recall.

4. Final Evaluation Results

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EVALUATION RESULTS
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📊 OVERALL METRICS (Weighted Average):
Accuracy: 0.9863 (98.63%)
Precision: 0.9874 (98.74%)
Recall:    0.9863 (98.63%)
F1-Score: 0.9865 (98.65%)

📊 OVERALL METRICS (Macro Average):
Precision: 0.9516 (95.16%)
Recall:    0.9884 (98.84%)
F1-Score: 0.9690 (96.90%)

📊 PER-CLASS METRICS:
Class 0 (Safe):
  Precision: 0.9988 (99.88%)
  Recall:    0.9856 (98.56%)
  F1-Score: 0.9921 (99.21%)
Class 1 (Scam):
  Precision: 0.9045 (90.45%)
  Recall:    0.9913 (99.13%)
  F1-Score: 0.9459 (94.59%)

📊 CONFUSION MATRIX:
          Predicted
          Safe  Scam
Actual Safe 4923   72
       Scam    6  682

True Negatives (TN): 4923 - Correctly predicted Safe
False Positives (FP): 72 - Incorrectly predicted Scam (Safe labeled as Scam)
False Negatives (FN): 6 - Incorrectly predicted Safe (Scam labeled as Safe)
True Positives (TP): 682 - Correctly predicted Scam

📊 DETAILED CLASSIFICATION REPORT:
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              precision    recall  f1-score   support

   Safe      0.9988      0.9856      0.9921     4995
   Scam      0.9045      0.9913      0.9459      688

 accuracy            0.9863     5683
 macro avg      0.9516      0.9884      0.9690     5683
weighted avg      0.9874      0.9863      0.9865     5683
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5. Error Analysis

During evaluation a small number of classification mismatches were observed.

False Positives (Safe labeled as Scam): 72

False Negatives (Scam labeled as Safe): 6

False positives were mostly caused by legitimate promotional or marketing posts that contained language like scam patterns.

False negatives occurred when scam messages used indirect wording or avoided common phishing indicators.

6. Key Improvements Implemented

Several improvements were introduced after initial testing to improve the system's reliability:

- Enhanced preprocessing to normalize slang and informal language
- Expanded scam training data with additional social media examples
- Integration of URL pattern analysis with AI-based text classification
- Optimization of the DistilBERT model for short-form social media text
- Risk score threshold tuning to reduce false positives

7. Conclusion

The evaluation results demonstrate that the Guardian Scam & Phishing Detection Module performs with high accuracy in detecting malicious messages and phishing attempts in social media content. The hybrid detection approach combining URL analysis and NLP-based language classification allows the system to detect both known malicious links and emerging scam patterns. With an overall accuracy of 98.63% and a scam recall rate of 99.13%, the module provides a reliable foundation for real-time scam detection within the Guardian browser extension.